

Part 1 Report — Reddit Data Analysis on ”Space Exploration”

Team 7

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1 Introduction

The goal of this project is to collect, analyze, and evaluate recent discussions about space exploration from social media platforms. This forms the groundwork for Part 2, where we’ll build a Knowledge Graph. We chose Reddit because it’s like the internet’s giant discussion room — filled with passionate space enthusiasts, curious learners, and the occasional flat-earthier.

We focused on extracting meaningful insights from posts related to space exploration, missions, planets, and celestial phenomena.

Tools & Technologies:

- Python 3 (executed in a code environment)
- PRAW API for accessing Reddit
- pandas, textblob, matplotlib, wordcloud for data analysis and visualization

- Time window: August 28 – October 5, 2025 (UTC)

2 Data Collected

Scope of collection:

- Main subreddit: **r/space** (focused on space exploration topics)
- Feed queried: **new** (latest 500 posts)
- Collected data: titles, bodies, URLs, scores, comment counts, and dates
- No separate comments fetched, but post interactions captured through scores and comment counts

Total records	500
Posts	500
Comments	(Not collected separately)
Time period	Aug 28 – Oct 5, 2025

Table 1: Data Summary

The dataset can be accessed here: [Reddit Data \(reddit_data.csv\)](#).

Basic descriptive statistics:

- Mean score (upvotes): ≈ 531
- Mean comments per post: ≈ 51

The data captures a good mix of space news and discussions — from asteroid flybys to breathtaking astrophotography. In short, it’s the kind of stuff that makes us wish we paid more attention in physics class.

3 Content Relevance

To check content relevance, we analyzed post titles for keywords such as *space*, *NASA*, *rocket*, *Mars*, *launch*, *star*, *moon*, *planet*, *asteroid*, *comet*, *satellite*. Each

post was assigned a score:

- 3 – Highly relevant (3+ keywords)
- 2 – Moderately relevant (2 keywords)
- 1 – Slight mention (1 keyword)
- 0 – Irrelevant (no keywords)

Relevance Score	Meaning	Share (%)
3	Highly relevant (3+ terms)	6.0
2	Moderately relevant (2 terms)	19.8
1	Slight mention (1 term)	37.8
0	Irrelevant (0 terms)	36.4

Table 2: Relevance Distribution

Most posts turned out to be at least somewhat relevant — with plenty of chatter about Mars, asteroids, and upcoming launches. Highly relevant ones usually referenced NASA updates or comet sightings. We also generated a word cloud (saved as `wordcloud.png`, not shown here) that highlighted frequent terms such as “NASA,” “Mars,” “Earth,” and “Space.”

For sentiment analysis, we used `TextBlob` to classify titles as positive, negative, or neutral. The average polarity was around 0.078 — slightly positive — meaning people are generally more thrilled than terrified about what’s happening beyond Earth.

The histogram shows most titles clustering around neutral, with a cheerful bias — reflecting excitement and curiosity rather than doom and gloom.

4 Link Structure Analysis

We analyzed URLs shared within posts to identify popular external sources, excluding internal Reddit links. Using Python’s `urlparse`, we extracted and ranked

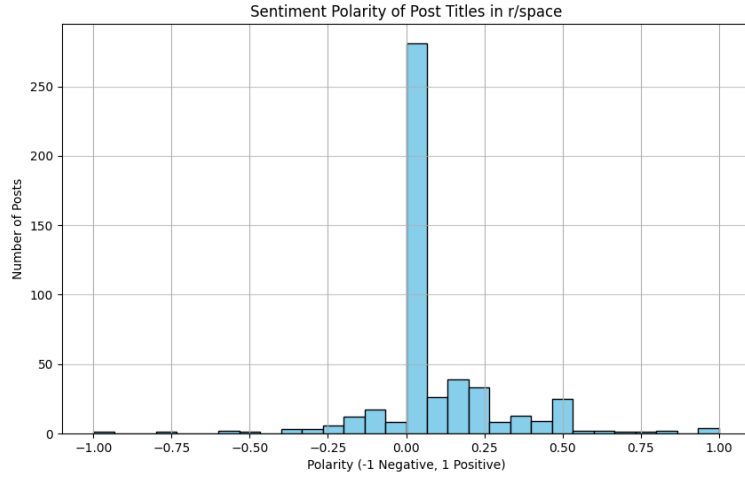


Figure 1: Sentiment Polarity Histogram of Post Titles

domains to see what kind of sites space fans are linking to.

Key observations:

- Around 40% of posts contain external links.
- Average links per post: ≈ 0.4 (some have none, some share multiple).

Rank	Domain	Mentions
1	i.redd.it	90
2	www.space.com	28
3	arstechnica.com	19
4	spacenews.com	16
5	www.nasa.gov	12
6	www.youtube.com	11
7	youtu.be	11
8	science.nasa.gov	10
9	www.esa.int	6
10	x.com	6

Table 3: Top 10 Most Shared External Domains

The results show a healthy mix — image hosting (i.redd.it), news portals (space.com, arstechnica.com), and official agencies (nasa.gov, esa.int). In short, Redditors aren't just posting memes — they're linking to credible, space-worthy content. Though yes, there's always that one post linking to a conspiracy YouTube video.

5 Evaluation of Methodology (Effectiveness)

To evaluate our approach, we assessed reliability, coverage, and relevance:

- Total posts collected: 500
- Posts with relevant keywords: 318 (approx. 64% precision)
- **Strengths:** The PRAW API made it easy to gather recent posts automatically. Keyword matching gave a quick relevance filter, and sentiment analysis helped gauge community mood.
- **Weaknesses:** Some genuinely relevant posts slipped through if they used uncommon terms (like “Artemis” without “NASA”). The Reddit API rate limits also capped us at 500 posts.
- **Overall:** The method worked well for a first pass — precise, recent, and surprisingly fun. With a few tweaks (like smarter NLP-based tagging), we could make the process even more accurate.

In conclusion, this approach gives a solid foundation for understanding how people discuss space exploration online — setting the stage for Part 2’s Knowledge Graph.

A Appendices

Sample data from [reddit_data.csv](#) (first few rows):

```
id,title,body,url,score,num_comments,created_utc
1nyib5f,"The best shot of Pleiades I've ever taken [OC]","Did a test run with the '
Gear: Nikon Z6 + TTArtisan 500mm f/6.3 Exposure: 124 × 120s @ f/7.3, ISO 3200 Mount
... (and more)
```

The complete dataset can be viewed and downloaded here: [Reddit Data \(Google Drive Link\)](#).

Direct link (hardcoded): https://drive.google.com/file/d/14-2vrf4zvgxA0K180Hf-M7mo28/view?usp=drive_link